

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA1001

COURSE NAME: Software Engineering

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Problem 1: Applying Scrum in a Telemedicine App Project

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Problem 1 - Applying Scrum in a Telemedicine App Project</title>

<style>
*{
  margin:0;
  padding:0;
  box-sizing:border-box;
  font-family:Arial, Helvetica, sans-serif;
}

body{
  background:#f4f6f9;
  padding:30px;
}

.container{
  max-width:1100px;
  margin:auto;
  background:white;
  padding:30px;
  border-radius:12px;
  box-shadow:0 0 15px rgba(0,0,0,0.15);
}

h1{
```

```
    text-align:center;
    color:#1565c0;
    margin-bottom:10px;
}

h2{
    margin-top:25px;
    color:#0d47a1;
    border-left:6px solid #1565c0;
    padding-left:10px;
}

p{
    line-height:1.8;
    margin-top:10px;
}

table{
    width:100%;
    border-collapse:collapse;
    margin-top:20px;
}

table th{
    background:#1565c0;
    color:white;
    padding:12px;
}

table td{
    border:1px solid #ccc;
    padding:12px;
    text-align:left;
}

ul{
    margin-left:25px;
    margin-top:10px;
}

li{
    margin:10px 0;
}

.card{
    background:#e3f2fd;
    border-left:6px solid #1565c0;
    padding:15px;
    margin-top:15px;
    border-radius:8px;
```

```
}
```

```
.footer{  
  margin-top:30px;  
  text-align:center;  
  color:#555;  
  font-weight:bold;  
}  
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h1>Problem 1: Applying Scrum in a Telemedicine App Project</h1>
```

```
<p>
```

A telemedicine mobile application project is experiencing frequent changes in requirements because the Product Owner continuously modifies the Product Backlog during Sprint Planning. This creates confusion among stakeholders, reduces team productivity, and affects Sprint Goals.

```
</p>
```

```
<h2>1. Scrum Roles and Responsibilities</h2>
```

```
<table>
```

```
<tr>
```

```
<th>Role</th>
```

```
<th>Responsibilities</th>
```

```
</tr>
```

```
<tr>
```

```
<td>Product Owner</td>
```

```
<td>
```

Defines product vision, prioritizes Product Backlog, communicates customer requirements, and accepts completed work.

```
</td>
```

```
</tr>
```

```
<tr>
```

```
<td>Scrum Master</td>
```

```
<td>
```

Facilitates Scrum events, removes impediments, ensures Scrum principles are followed, and protects the team from unnecessary requirement changes.

```
</td>
```

```
</tr>
```

```
<tr>
```

<td>Development Team</td>
<td> Designs, develops, tests, and delivers a working product increment during each sprint. </td>
</tr>
<tr>
<td>Stakeholders</td>
<td> Provide business feedback, requirements, and validate product functionality. </td>
</tr>
</table>

<h2>2. Root Cause Analysis</h2>

- Frequent backlog modifications during Sprint Planning.
 - Lack of Product Backlog refinement before sprint starts.
 - Changing customer requirements without impact analysis.
 - Poor communication between stakeholders and Scrum Team.
 - No stable Sprint Goal established.
 - Improper requirement prioritization.

<h2>3. Improved Backlog Prioritization Plan</h2>

- Conduct regular Product Backlog Refinement meetings.
 - Freeze Sprint Backlog once Sprint Planning is completed.
 - Prioritize using the MoSCoW Method:

 - Must Have
 - Should Have
 - Could Have
 - Won't Have- Estimate user stories using Story Points.
- Define clear Acceptance Criteria for every backlog item.
- Review priorities before each sprint instead of during the sprint.

<h2>4. Retrospective Improvement Suggestions</h2>

```
<div class="card">
<ul>
<li>Improve stakeholder communication.</li>
<li>Conduct backlog grooming every sprint.</li>
<li>Avoid changing sprint scope after Sprint Planning.</li>
<li>Maintain Definition of Ready (DoR).</li>
<li>Track Sprint Velocity for better planning.</li>
<li>Identify blockers early and resolve them quickly.</li>
<li>Encourage continuous feedback from the development team.</li>
</ul>
</div>
```

<h2>Conclusion</h2>

```
<p>
By clearly defining Scrum roles, stabilizing requirements through proper backlog refinement, using MoSCoW prioritization, and implementing retrospective action items, the telemedicine project can improve productivity, reduce requirement churn, and deliver high-quality software increments successfully.
</p>
```

```
<div class="footer">
Scrum Project Analysis - Telemedicine Mobile App
</div>
```

```
</div>
```

```
</body>
</html>
```